ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Red Lion / N-Tron 700 Series Managed Industrial Ethernet Switch · Component type: **Network Device (NDR) - managed industrial Ethernet switch (legacy)** · Target SL = SL 3

Where to work	The switch **web UI** (default http://192.168.1.201, **admin/admin**) - Administration (password, SNMP, firmware), Ports (Port Security / MAC filtering), VLAN, Bridging/N-Ring, RSTP; the serial **console CLI**; **SNMP** (prefer v3). Monitored by **N-View** (see the N-View sheet). No HTTPS/SSH on this legacy platform.
Ports / paths	**HTTP 80** (web mgmt, cleartext - no HTTPS); **Telnet 23** / serial console (no SSH); **SNMP UDP 161** (v1/v2/v3) + traps 162; **TFTP 69** (firmware/config); N-View L2 multicast. Default IP **192.168.1.201**. Management is cleartext - restrict to a dedicated management VLAN.
Firmware / platform	N-Tron **700 Series** (708TX / 709FX / 710FX2 / 716 / 708M12 etc.). Legacy: no HTTPS/SSH/802.1X/MACsec/secure boot. Confirm model + firmware in the web UI; track Red Lion / CISA advisories and apply firmware. Device auth is **Port Security MAC filtering** (MAC-based).
CLI syntax (700 — web/console, not IOS)	This legacy N-Tron has a limited **serial-console CLI** and **Telnet** (no SSH) — it is **not** Cisco IOS, so IOS `show` verbs do not apply. Capture evidence in the **web UI** (Administration / Ports / VLAN / Bridging) and via **SNMP** (prefer v3), and pull the config over **TFTP** (`copy` is web/TFTP-driven, not an IOS command line). Because HTTP/Telnet/SNMPv1-2 are cleartext, do the review from the dedicated management VLAN only. Confirm model/firmware in the web UI.
How to read	Default read-only (web/CLI read-only account, read-only SNMP). Each row = native step-by-step + command-level external-tool validation + pass, with honest Partial/Compensating/N-A where the legacy platform has no native control. Write/exploit tools = isolated env + written authorization.

Session setup & evidence handling

Do this BEFORE running any command so the evidence is complete, attributable, and protected:
1. Start a full session log first. PuTTY: Session › Logging → select **All session output** → set **Log file name** (browse to the encrypted evidence folder) → under “if the log file already exists” pick **Always append** → tick **Include header** and (Logging › Options) **Flush log frequently**; put `&Y&M&D_&T` in the name for an auto timestamp. (SecureCRT: File › Log Session; Tera Term: File › Log; macOS/Linux: `script -a <file>.log` or `ssh user@host | tee <file>.log`.)
2. Name the file to the evidence convention (one file per device): `<site>_<zone>_<assetTag>_<vendor>_<model>_<mgmtIP>_<YYYYMMDD-HHMM>_<testerInitials>.log`.
3. Disable paging so full output is captured (it resets each login): web-first platform: capture the web UI with the browser (print-to-PDF / screenshots) + read-only SNMPv3; for the serial console use PuTTY logging (there is no IOS-style pager).
4. Self-document the log - capture identity first from the web UI (Help / About: model + firmware), so the file proves what/when/who.
5. Close out: restore paging, stop the log (PuTTY closes it when the session ends), and verify the file opened/saved.
6. Protect the evidence file. It contains running-config and may hold hashes, SNMP community strings, or keys: store **only** on the encrypted assessment share, follow the project data-classification/retention policy, redact secrets before sharing, and **never** email it in cleartext. **Do not save the device password in the PuTTY/SecureCRT saved session** - authenticate interactively or via the approved password manager.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.1	Human user I&A	Confirm the web UI + CLI require login and the **default admin/admin is changed** (Administration › Password). The 700 series has a limited user model (admin, and a lower-privilege user on some models) - use it; there is no per-user directory. No unauthenticated management path. **Test:** browse `http://<sw>` / console without creds -> login required; default admin/admin rejected. **Ext. tools:** `curl -s -o /dev/null -w "%{http_code}" http://<sw>/` (expect auth); attempt admin/admin (authorized) -> must fail after change.	Web + CLI authenticate; default password changed; no anonymous management.	Limited user model (residual).

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.2	Device & software-process I&A	Device authentication is **Port Security - MAC address filtering** (Ports › Security): lock each access port to the allowed MAC(s) so unknown devices are dropped. Note MAC filtering is spoofable (no 802.1X on this platform) - compensate with physical port control + monitoring. **Test:** connect an unlisted MAC to a secured port -> traffic dropped / port action taken. **Ext. tools:** Web (Port Security config); rogue-MAC test on a lab port; Wireshark.	Port Security MAC filtering on access ports; unknown MACs dropped (spoofable residual noted).	No 802.1X - Partial.
SR/CR 1.3	Account management	Manage the switch admin/user credentials as controlled secrets (least privilege, no sharing, defaults removed); there is no central directory. Document custody. **Ext. tools:** Web (accounts); credential-custody record.	Admin/user credentials controlled/least-privilege; defaults removed.	Compensating.
SR/CR 1.4	Identifier management	Confirm a unique device identity (IP - change from the 192.168.1.201 default, name, MAC/serial) per switch; no reuse. **Ext. tools:** Web/CLI device info; `nmap -sU -p 161 --script snmp-info <sw>`; inventory.	Unique device identity; default IP changed; no reuse.	
SR/CR 1.5	Authenticator management	Set a strong, non-default admin (and user) password; store it in the vault. Note the legacy platform has limited password-policy enforcement - capture what exists and compensate. Change SNMP community strings from public/private. **Ext. tools:** Web (password set; SNMP communities changed); policy capture.	Strong non-default passwords + SNMP communities; policy captured (weak residual noted).	Partial (legacy).
SR/NDR 1.6	Wireless access management	**N/A.** Wired switch with no native wireless interface. **Ext. tools:** Confirm no WLAN interface.	N/A - justification recorded (wired network device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A / limited.** The 700 series web/CLI have no TLS/SSH (HTTP/Telnet only). Record N/A; the only crypto available is **SNMPv3** (SHA/AES) - use it. Provide management confidentiality via a segmented management VLAN + jump host. **Ext. tools:** Confirm no HTTPS/SSH (`nmap -p 443,22 <sw>` closed); SNMPv3 auth/priv config.	N/A/limited PKI (no HTTPS/SSH); SNMPv3 crypto; compensating management-VLAN.	N/A/Partial.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	Confirm the web/CLI login masks password entry (no cleartext echo). Note HTTP/Telnet carry the login in cleartext on the wire -> confine to the management VLAN (see 1.13/3.1). **Ext. tools:** Visual verification; Wireshark (login cleartext -> management-VLAN only).	Authenticator obscured at entry; cleartext login confined to the management VLAN.	
SR/CR 1.11	Unsuccessful login attempts	The 700 series has **no account lockout**, but it raises an **Authentication Error SNMP trap** on failed access - forward it to the SIEM; compensate with the management-VLAN restriction + monitoring. **Ext. tools:** Web (Authentication-Error trap enabled -> SIEM); confirm no native lockout.	Auth failures trapped/forwarded; no native lockout (residual + compensating).	Partial/Compensating.
SR/CR 1.12	System use notification	No native banner; enforce the approved use notice on the engineering-workstation host (GPO) and any NMS (compensating). **Ext. tools:** EWS GPO; NMS banner review.	Use notice at EWS/NMS (compensating at switch).	Compensating.
SR/NDR 1.13	Access via untrusted networks	Because management is cleartext (HTTP/Telnet/SNMP), confine it to a dedicated **management VLAN** reachable only from the trusted zone; disable management on user ports; remote admin via VPN/jump host; no management from untrusted networks. **Test:** from an untrusted VLAN, reach 80/23/161 -> must fail. **Ext. tools:** `nmap -Pn -p 80,23,161 <sw>` from the untrusted side (expect filtered); management-VLAN + firewall review.	Management confined to a trusted management VLAN; untrusted access denied; remote via VPN.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.1	Authorization enforcement	Confirm least privilege via the switch roles (admin vs user where available) - a user-level login cannot change configuration; restrict who has the admin credential. **Test:** as a user-level login, attempt a config change -> denied. **Ext. tools:** Web (role/permission review); user-level negative test.	Least privilege via admin/user roles; config change denied to user.	Limited roles.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR 2.3	Use control for portable/mobile devices	Control the engineering laptop and any USB/config media; removable-media policy on the EWS; only authorized firmware/config loaded (TFTP source controlled). **Ext. tools:** EWS removable-media policy; TFTP-source control.	Portable/engineering media controlled.	
SR/NDR 2.4	Mobile code	**N/A / limited.** No general-purpose app-hosting/mobile-code runtime on this switch; confirm only the vendor firmware + web UI run (no user script path). **Ext. tools:** Confirm no scripting/app-hosting; web content vendor read-only.	No mobile-code runtime; only vendor firmware runs.	NDR-specific.
SR/CR 2.5	Session lock	Confirm the web/CLI session times out on inactivity (re-login required); enforce the host OS lock on the management workstation. **Ext. tools:** Web/CLI timeout config; host inactivity-lock GPO.	Sessions time out on inactivity; re-auth required.	
SR/CR 2.6	Remote session termination	Confirm management sessions can be ended and idle sessions close; the switch bounds concurrent management connections. **Ext. tools:** Web/CLI session review; Wireshark (session teardown).	Sessions terminable; concurrency bounded.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	The 700 series has limited on-box logging and **no syslog on older firmware**; forward **SNMP traps** (Auth Error, Cold/Warm Start, Link Up/Down) to the SIEM and use N-View/an OT IDS for change/anomaly capture. Confirm the firmware's syslog capability if present. **Ext. tools:** Web (trap stations -> SIEM); N-View/OT IDS; confirm syslog if available.	Security events trapped/forwarded; off-device monitoring captures changes.	Limited native audit - Compensating.
SR/CR 2.9	Audit storage capacity	On-box logging is limited - forward SNMP traps to the SIEM for retention and confirm the SIEM alerts if the trap/monitor feed goes silent so events are not lost. **Ext. tools:** SIEM retention + source-silent alert; confirm trap delivery.	Events offloaded/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	Confirm the switch time is synced (SNTP where supported) so trap/log timestamps are accurate and correlate; else the NMS/SIEM applies receive-time. **Ext. tools:** Web (SNTP config); Wireshark `ntp`.	Synchronized timestamps where supported; NMS receive-time otherwise.	Partial.
SR/CR 2.12	Non-repudiation	Attribution is limited (shared admin account); compensate by restricting the admin credential to named individuals + logging management access at the jump host / NMS. **Ext. tools:** Jump-host/NMS access logs; credential-custody record.	Access attributable via jump host/custody (shared-account residual noted).	Compensating.
NDR 2.13	Physical & diagnostic interfaces	Control physical interfaces: locked cabinet; secure the serial **console** and the reset; shut/disable unused ports; keep the DIN-rail unit sealed. There is no USB config port on most 700 models. **Ext. tools:** Physical inspection; Web/CLI (unused ports shut/disabled).	Console/reset/unused-ports controlled; enclosure sealed.	NDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 3.1	Communication integrity	Management is **cleartext** (HTTP/Telnet/SNMPv1-2, no HTTPS/SSH) - so integrity/confidentiality of management come from a **segmented management VLAN** + jump host; use **SNMPv3** (auth+priv) and disable v1/v2; data-plane integrity via VLAN isolation. No cleartext management across a zone boundary. **Test:** confirm management stays in the management VLAN; SNMPv3 in use. **Ext. tools:** Wireshark (HTTP/Telnet/SNMP; confirm in-zone + v3); `nmap -sU -p 161 <sw>`.	Management contained to a management VLAN; SNMPv3; cleartext not crossing the boundary.	No HTTPS/SSH - Compensating.
SR/NDR 3.2	Protection from malicious code	Confirm only Red Lion/N-Tron firmware runs. **No signed-firmware/secure-boot guarantee** on this legacy platform - verify firmware provenance from Red Lion and control the TFTP update path (2.3/3.10). **Ext. tools:** Web (firmware version); Red Lion provenance/hash; Red Lion/CISA advisory.	Only vendor firmware runs; provenance verified; no code path.	NDR-specific/Partial.
SR/CR 3.3	Security functionality verification	After any change re-verify: default password changed, SNMPv3/no-v1-v2, Port Security, VLANs, RSTP/N-Ring, management-VLAN restriction, and firmware, against the hardening baseline; back up + compare the switch config. **Ext. tools:** Config backup/compare (TFTP); `nmap -p 80,23,443,22,161 <sw>` (hardened state).	Config matches baseline; unauthorized change detected.	
SR/CR 3.4	Software & information integrity	Confirm software & information (config/program) integrity; unauthorized change prevented/detected. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Software & information integrity verified individually.	
SR/CR 3.5	Input validation	Confirm the switch tolerates malformed management/protocol frames without instability. **ISOLATED only:** fuzz the web/SNMP/RSTP stacks on a bench switch and watch stability. **Ext. tools:** Protocol fuzzer (ISOLATED) on a bench switch; Wireshark; watch CPU/stability.	No instability on malformed input.	Do NOT fuzz an in-service switch.
SR/CR 3.6	Deterministic output	**N/A.** A switch does not drive a physical process output. Confirm the configured link/ring failure behavior (N-Ring healing, RSTP convergence) is deliberate. **Ext. tools:** Web (N-Ring/RSTP config); ring-break test (lab) -> healing within spec.	N/A - justification recorded (network device); ring/RSTP behavior verified.	N/A
SR/CR 3.7	Error handling	Confirm graceful error handling without info leak; review the switch event/status log; err-disable/port behavior deliberate. **Ext. tools:** Web/CLI (event log / port status review).	Graceful error handling; no info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** Management sessions are cleartext (HTTP/Telnet); SNMPv3 sessions are integrity-protected. Contain the rest by the management VLAN. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (SNMPv3 vs cleartext HTTP/Telnet).	SNMPv3 integrity-protected; cleartext management contained.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since native logging is limited/off-device, protect the SIEM/NMS store and restrict who can clear the switch event log (admin-only); protect the config backups (they contain SNMP/credential data). **Ext. tools:** SIEM/NMS access review; confirm log-clear is admin-only; secure config backups.	Audit records protected from unauthorized read/modify/delete.	
NDR 3.10	Support for updates	Update firmware via the web UI / TFTP with firmware from Red Lion; verify version/provenance; plan a maintenance window with rollback to the prior version. No on-device signature verification - provenance discipline is the control. **Ext. tools:** Web/TFTP (firmware update); Red Lion provenance/version.	Controlled updates from a verified source, with rollback.	NDR-specific/Partial.
NDR 3.11	Physical tamper resistance/detection	Keep the switch in a locked cabinet; control the console/reset; the event log + SNMP traps (Cold/Warm Start) provide tamper/restart evidence; seal where required. **Ext. tools:** Physical inspection (lock/seal); event-log / Cold-Start trap review.	Cabinet locked/sealed; interfaces controlled; restarts logged.	NDR-specific.
NDR 3.12	Provisioning product-supplier roots of trust	**N-A/Partial at device.** No signed-boot supplier root of trust. Record the gap; compensate via firmware-provenance discipline (3.10), physical control, and segmentation. **Ext. tools:** Confirm no signed-firmware/secure-boot root; provenance review.	No supplier RoT; compensating controls recorded.	Partial/N-A.
NDR 3.13	Provisioning asset-owner roots of trust	**N-A/Partial at device.** No asset-owner certificate/PKI provisioning (no TLS). Record the gap; asset-owner trust is enforced at the management-VLAN/network layer. **Ext. tools:** Confirm no certificate/PKI capability; management-VLAN review.	No asset-owner RoT; compensating controls recorded.	Partial/N-A.
NDR 3.14	Integrity of the boot process	**N-A/Partial at device.** No verified/secure boot. Record the gap; mitigate via firmware-provenance discipline (3.10), physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability; firmware version.	No verified boot; compensating controls recorded.	Partial/N-A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/RMA, reset the switch to factory defaults (clears config, credentials, SNMP communities) and remove any saved config/backups so nothing sensitive is recoverable. **Ext. tools:** Web/CLI (factory reset); confirm saved configs removed.	Config, credentials, and communities purged on decommission.	
SR/CR 4.3	Use of cryptography	**N/A / limited.** No management TLS/SSH; the only crypto is SNMPv3 (SHA/AES). Rely on the management VLAN + jump host for confidentiality; disable SNMP v1/v2 and unused services. **Ext. tools:** Confirm SNMPv3 auth/priv; v1/v2 disabled; management-VLAN review.	SNMPv3 crypto; no management TLS; compensating segmentation.	N/A/Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 5.1	Network segmentation	Strong point: use **802.1Q VLANs** to isolate zones and a dedicated **management VLAN**; put unused ports in an unused VLAN and shut them; confirm the VLAN/port map matches the zone model. **Ext. tools:** Web (VLAN/port map); `nmap` across VLANs to prove isolation.	Effective VLAN segmentation; management VLAN isolated; unused ports shut.	
SR/NDR 5.2	Zone boundary protection	The 700 series is a Layer-2 switch - boundary filtering is at the connected firewall/router + the management-VLAN restriction + Port Security; confirm only required flows cross the zone boundary. **Ext. tools:** `nmap` at the boundary; firewall/VLAN-ACL review.	Boundary enforces least-flow (via firewall + VLAN + Port Security).	
SR/NDR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Network switch with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services on the switch.	N/A - justification recorded.	N/A
SR 5.4	Application Partitioning	**N/A.** Single-purpose device; no separable applications to partition (handled at the host/SCADA layer).	N/A - justification recorded (not assessed as a control).	N/A

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 6.1	Audit log accessibility	Confirm the switch event log + SNMP traps are readable/exportable and forwarded to the SIEM/NMS; N-View + a passive OT IDS on the segment provide continuous monitoring (limited native syslog on legacy firmware). **Ext. tools:** Web/CLI log export; SNMP traps -> SIEM; N-View / passive OT IDS.	Events accessible/exportable and monitored in near-real-time.	Limited native syslog - Compensating.
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 7.1	Denial-of-service protection	Confirm resilience: **N-Ring/N-Link** ring redundancy (~30 ms heal) + RSTP, IGMP snooping, and Port Security bound broadcast/flood impact; management-VLAN restriction limits the mgmt-plane attack surface. Monitor under load. **Ext. tools:** Web (N-Ring/RSTP/IGMP status); ISOLATED flood test on a bench; watch stability.	Ring/RSTP redundancy + Port Security keep the switch forwarding under fault/load.	
SR/CR 7.2	Resource management	Confirm bounded resources (MAC table limits, connection/mgmt limits) so a flood cannot exhaust the switch; IGMP snooping controls multicast load. **Ext. tools:** Web (MAC table / IGMP status); bench load test.	Resources bounded; no exhaustion-based loss of forwarding.	
SR/CR 7.3	Control system backup	Confirm the switch config is backed up (versioned, offline, via TFTP) and a restore is tested on a spare switch for a known-good rebuild. **Ext. tools:** Config backup (TFTP); restore test (spare switch).	Config backup current; restore/rebuild tested.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: change default admin/admin + SNMP communities, disable **Telnet** and unused services, use **SNMPv3** (disable v1/v2), confine management to the management VLAN, enable Port Security, shut unused ports, and patch firmware. **Ext. tools:** Web (services/SNMP/Port Security config); `nmap -sTU -p 23,80,161,69 <sw>` (only required reachable).	Least functionality; Telnet/v1-v2/unused services disabled; hardened.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Confirm the switch (model, firmware, ports) is inventoried so vulnerabilities track against Red Lion / CISA advisories. **Ext. tools:** `nmap -sU -p 161 --script snmp-info <sw>`; web inventory; Red Lion/CISA feeds.	Accurate model/firmware inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
Web UI / console CLI	Native config review (all FRs)	Browse http://<sw> (change default admin/admin) or the serial console; review Administration (password/SNMP/firmware), Port Security, VLAN, N-Ring/RSTP; back up the config via TFTP.	Read/authorized; unlock only for change tests.
nmap	Port/service exposure & segmentation (1.x, 3.1, 5.x, 7.6, 7.8)	`nmap -sTU -p 23,80,161,69,443,22 <sw>`; `nmap -sU -p 161 --script snmp-info`; scan per VLAN; confirm 443/22 closed and Telnet disabled.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & SNMP (1.10, 3.1, 3.8)	Confirm management (HTTP/Telnet/SNMP) stays in the management VLAN; SNMPv3 (not cleartext v1/v2); inspect N-Ring/RSTP.	Passive; SPAN/tap only.
curl	Web-auth check (1.1)	`curl -s -o /dev/null -w "%{http_code}" http://<sw>/` - confirm auth required; confirm default admin/admin fails after change.	Read-only; authorized.
SNMP tools (snmpwalk)	SNMP config & version (1.2, 4.3, 7.8)	`snmpwalk -v3 ...` - confirm v3 works and v1/v2 (public/private) are disabled/changed.	Read-only; authorized.
Rogue-MAC test device	Port Security (1.2)	Connect an unlisted MAC to a secured lab port; confirm the MAC filter drops it / takes the configured action.	Coordinate; lab port.
Protocol fuzzer / traffic generator	Input validation & DoS resilience (3.5, 7.1)	Malformed/high-rate web/SNMP/RSTP on a bench switch; ring-break/flood test; watch stability.	ISOLATED bench; never in-service.

Tool / method	Verifies	How to use — steps / example	Constraints
SNMP trap receiver / SIEM + passive OT IDS	Monitoring (2.8, 6.x)	Receive the switch traps (Auth Error/Cold-Start/Link) at the SIEM; N-View / OT IDS for change/anomaly.	Coordinate with SOC.
Red Lion / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track N-Tron 700-series advisories and required firmware.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.